

Juan David Muñoz Bolaños

Optical Metrology Engineer

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Summary

Physicist and optical engineer finishing a PhD in June 2026, with four years of hands-on lab work building, aligning, and calibrating precision optical measurement systems from scratch. Before that, I spent a year installing and commissioning automated inspection machines at customer sites in industry. I'm comfortable with PC and control technology, used to digging into technical problems systematically, and I enjoy direct contact with the people who use the systems I work on. Based in Tirol, fully authorised to work in Austria, and happy to travel within Austria and nearby countries.

Research/Professional Experience

Jan 2022 to Jun 2026 **Med. Univ. Innsbruck**

Innsbruck, Austria

Optical Metrology Engineer | PhD Researcher

Building & Calibrating Precision Optical Systems: Built complex optical measurement systems from scratch in the lab, aligned and calibrated them by hand, and kept them running. Worked with scientific cameras, piezo stages, galvo scanners, deformable mirrors and DAQ hardware. Did the routine maintenance, alignment, and fault-finding myself.

PC & Control Software: Wrote the control software in Python, LabVIEW (Real Time, FPGA) and C++ for automated measurement, calibration and data acquisition. Set up the PC-based measurement workstations and tracked down hardware-software issues when something didn't behave.

Troubleshooting & Documentation: Planned and ran system-level tests, did root cause analysis on optical and electronic faults, and wrote up clear technical documentation so others could pick up the work.

Mar 2018 to Feb 2019 **INTECOL S.A.S.**

Medellín, Colombia

Machine Vision Engineer (Field Service)

On-Site Commissioning & Calibration: Installed, calibrated and commissioned automated optical inspection machines directly at customer production sites (Halcon, Cognex VisionPro). Verified each system against acceptance criteria and got customer sign-off before leaving.

Service, Training & Customer Support: Handled on-site repairs, retrofits and end-user training. Worked closely with production, mechanical design and maintenance teams, did root cause analysis on live lines, and pushed through small improvements that made the machines more reliable.

Technical Competencies

Metrology & Optical Measurement: Optical alignment and calibration, interferometry, wavefront sensing, phase retrieval, illumination design (bright-field, dark-field, ring-light), uncertainty analysis.

PC & Control Technology: Python (NumPy, SciPy), C++ (OpenCV), LabVIEW (Real Time, FPGA), MATLAB, DAQ cards (National Instruments), microcontroller programming (Arduino, Code Composer Studio), Git.

Machine Vision & Inspection: Halcon, Cognex VisionPro, OpenCV, automated defect detection, industrial camera integration (Basler, Andor, Hamamatsu).

Hardware & Hands-on Skills: Opto-mechanical design (SolidWorks, FreeCAD), galvo scanners, piezo stages (PI), deformable mirrors, laser systems, fiber optics.

Foundational PCB design (Proteus). Manual workshop skills for instrument assembly and alignment.

Education

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>Doctor of Philosophy in Optical Metrology</i>	<i>Innsbruck, Austria</i>
2019 to 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	<i>Jena, Germany</i>
2012 to 2018	University of Cauca <i>Bachelor of Science in Physics Engineering</i>	<i>Popayán, Colombia</i>

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

• Mountain Sports (Biking, Hiking, Skiing) • Bachata Dancing • Harmonica
• Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte <i>Medical University of Innsbruck</i> monika.ritsch-marte@i-med.ac.at	PD Dr. Christoph Krafft <i>Leibniz IPHT</i> christoph.krafft@leibniz-ipht.de
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